# Right Triangles & Trigonometry

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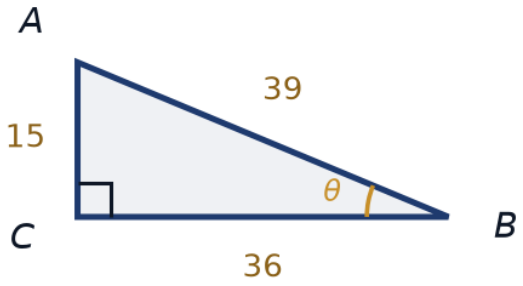
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In $\Delta ABC$ studying proving triangles congruent, $m\angle A = 31^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

**2** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In $\Delta ABC$ studying corresponding parts of congruent triangles, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.




**3** [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 2 · MediumIn a right triangle $ABC$ where angle $C = 90^\circ$, $\cos(A) = \frac{5}{13}$. What is the value of $\tan(A)$?

**4** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a right triangle, acute angle $\theta$ satisfies $\sin\theta = \frac{5}{13}$. Find the exact values of $\cos\theta$, $\tan\theta$, and $\sec\theta$.

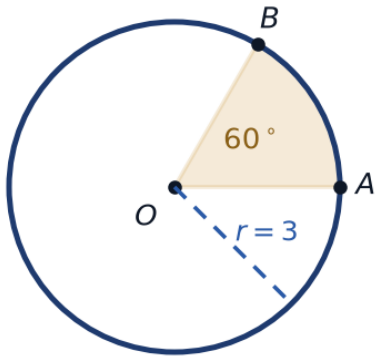
**5** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In right triangle $ABC$ with $C = 90^\circ$, $a = 5$, and $c = 13$. What is the value of $\sin A + \cos A$?

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**6** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 3 · AdvancedThe equation of a circle in the $xy$-plane is $x^2 - 8x + y^2 + 6y = 24$. What are the coordinates of the center and the radius of the circle?




**7** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 4 · SynthesisThe equation of a circle in the $xy$-plane is $x^2 + y^2 - 10x + 6y - 15 = 0$. What is the length of the diameter of the circle?

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsA cone-shaped storage tank has a base radius of 4 feet and a height of 9 feet. What is the volume of the tank in cubic feet, rounded to the nearest tenth?

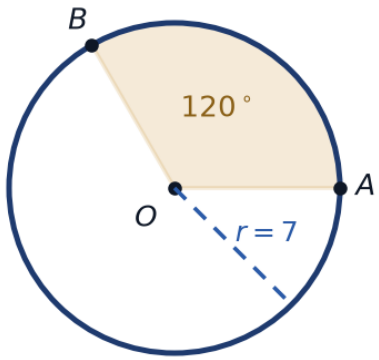
**9** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Simplify the expression $\left(\frac{16x^8y^{-4}}{81z^{12}}\right)^{-3/4}$ and write your answer using only positive exponents.

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**10** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.




**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 3 · AdvancedIn the system of equations below, $k$ is a constant. For what value of $k$ does the system have no solution? $\{kx - 6y = 15; 4x - 3y = 9$

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A right triangle has opposite $3$ and hypotenuse $5$. Find $\sin\theta$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Evaluate the complex expression $\frac{(3-2i)^2}{1+i}$ and write the result in standard form $a + bi$.

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle $XYZ$, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of $XZ$ to 3 significant figures and (b) the area of triangle $XYZ$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

A right triangle has $\tan\theta = \frac{3}{4}$. Find $\sin\theta$.

**16** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 10.01 Coordinate Geometry

If $\cos(2x + 10^\circ) = \sin(3x - 15^\circ)$, find the value of $x$ and evaluate $\tan(x + 16^\circ)$.

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**17** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 11.01 Probability

Which of the following expressions is equivalent to $\frac{1-\sin^2\theta}{\cos\theta}$ for all values of $\theta$ where the expression is defined?

**18** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

If $\cos\theta = -\frac{4}{5}$ and $\theta$ lies in Quadrant II, what is the value of $\tan\theta$?

**19** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

If $A = (2,\ -1; 3, 4)$ and $B = (5;\ -2)$, what is the matrix product $AB$?

**20** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

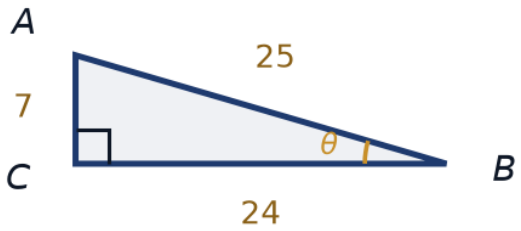
Find the hypotenuse of a right triangle with legs 7 and 24.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In $\Delta ABC$ studying angles of triangles, $m\angle A = 54^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?




**a.** $81^\circ$

**b.** $91^\circ$

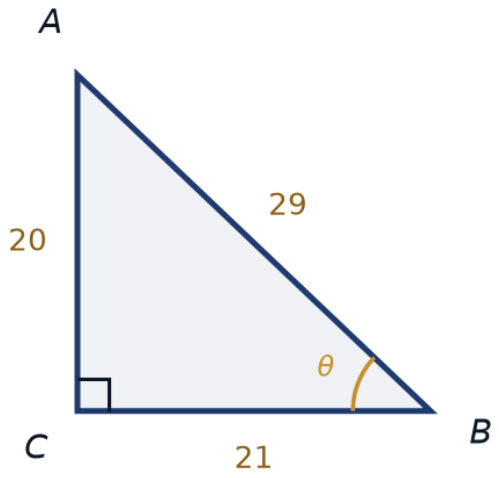
**c.** $73^\circ$

**d.** $126^\circ$

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**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In $\Delta ABC$ studying review: identifying congruent triangles, $m\angle A = 30^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?




**a.** $105^\circ$

**b.** $115^\circ$

**c.** $97^\circ$

**d.** $150^\circ$

**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In $\Delta ABC$ studying isosceles and equilateral triangles, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

**a.** $95^\circ$

**b.** $105^\circ$

**c.** $87^\circ$

**d.** $147^\circ$

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right triangle $ABC$, the measure of $\angle C$ is $90^\circ$. If $\sin A = \frac{x+1}{13}$ and $\cos B = \frac{3x-7}{13}$, what is the value of $\tan A$?

**a.** $\frac{5}{12}$

**b.** $\frac{12}{5}$

**c.** $\frac{5}{13}$

**d.** $\frac{12}{13}$

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Right triangle legs $3$ and $4$: hypotenuse:

**a.** 3

**b.** 6

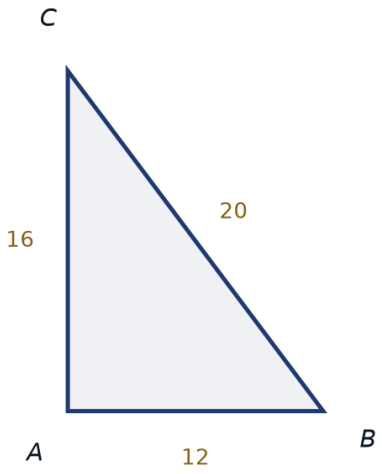
**c.** 7

**d.** 5

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**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A right triangle has legs 12 and 16 and hypotenuse 20. What is its area?




**a.** 48

**b.** 106

**c.** 96

**d.** 192

**27** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A right circular cone has a base radius of $5$ and a height of $12$. What is the lateral area of the cone?

**a.** $60\pi$

**b.** $65\pi$

**c.** $90\pi$

**d.** $130\pi$

**28** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right circular cylinder has a radius of 5 cm and a height of 10 cm. What is the total surface area of the cylinder?

**a.** $150\pi$ cm$^2$

**b.** $250\pi$ cm$^2$

**c.** $350\pi$ cm$^2$

**d.** $500\pi$ cm$^2$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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**29** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Two equilateral triangles have side lengths in the ratio $4:9$. What is the ratio of their altitudes?

**a.** $2:3$

**b.** $16:81$

**c.** $8:27$

**d.** $4:9$

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In right $\Delta ABC$ with right angle at $C$, altitude $CD \perp AB$. If $AD = 3$ and $DB = 12$, what is the length of leg $AC$?

**a.** $3\sqrt{5} \approx 6.71$

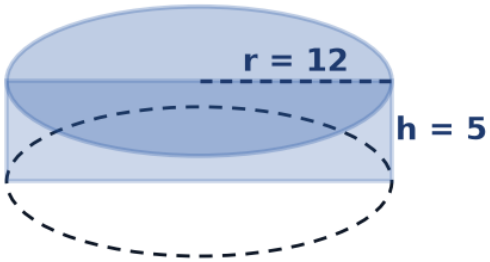
**b.** $6$

**c.** $15$

**d.** $6\sqrt{5} \approx 13.42$

**31** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?




**a.** $100\pi$

**b.** $\frac{400\pi}{3}$

**c.** $\frac{500\pi}{3}$

**d.** $200\pi$

**32** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In right $\Delta ABC$ with right angle at $C$, altitude $CD$ is drawn perpendicular to hypotenuse $AB$. If $AD = 4$ and $DB = 9$, what is the length of altitude $CD$?

**a.** $6.5$

**b.** $6$

**c.** $5.2$

**d.** $7.2$

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**33** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In $\Delta JKL$ with right angle at vertex $K$, an altitude from $K$ meets hypotenuse $JL$ at point $M$. Given that segment $JM = 5$ and $ML = 15$, determine the length of leg $JK$.

**a.** $5\sqrt{3}$

**b.** $10\sqrt{3}$

**c.** 10

**d.** $5\sqrt{5}$

**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In right $\Delta ABC$ with $\angle C = 90^\circ$, $AC = 6$, $BC = 8$, and hypotenuse $AB = 10$. The angle bisector of $\angle A$ intersects side $BC$ at point $D$. What is the exact length of segment $CD$?

**a.** 4

**b.** 2.5

**c.** 3

**d.** 3.5

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In right $\Delta ABC$, altitude $CD$ to hypotenuse $AB$ has length $12$. If segment $AD = 9$, what is the length of hypotenuse $AB$?

**a.** 16

**b.** 20

**c.** 21

**d.** 25

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Two similar triangles have a linear scale factor of $3:7$. If the perimeter of the smaller triangle is $24$, what is the perimeter of the larger triangle?

**a.** 48

**b.** 72

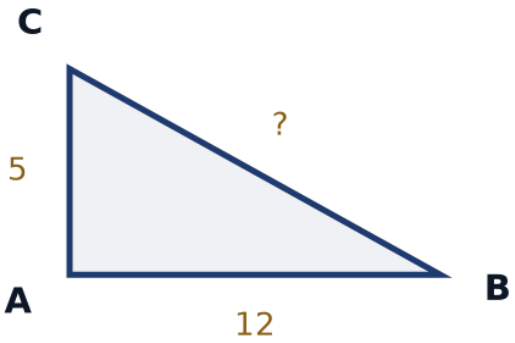
**c.** 130.67

**d.** 56

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a right triangle, the legs are 5 and 12. What is the hypotenuse?




**a.** 13

**b.** 15

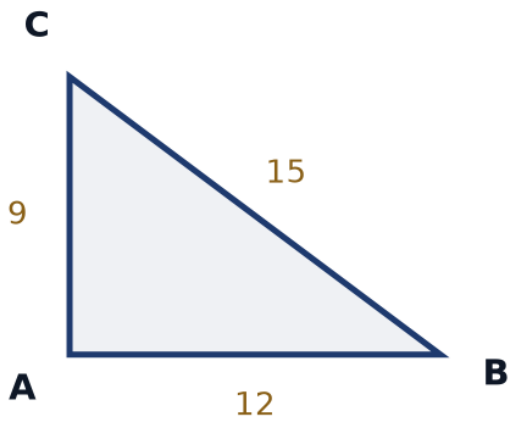
**c.** 17

**d.** $\sqrt{119}$

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Lines & Angles

In a right triangle, the legs have lengths 9 and 12. What is the area of the triangle?




**a.** 27

**b.** 54

**c.** 72

**d.** 108

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**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Triangles & Congruence

All equilateral triangles are equiangular; this triangle is equilateral, so it is equiangular. This is an example of:

**a.** Inductive reasoning

**b.** A logical fallacy

**c.** Circular reasoning

**d.** Deductive reasoning

**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

All right angles measure $90^\circ$; $\angle A$ is a right angle, so $\angle A = 90^\circ$. This is an example of:

**a.** A logical fallacy

**b.** Inductive reasoning

**c.** Circular reasoning

**d.** Deductive reasoning

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Polygons & Quadrilaterals

A student folds several isosceles triangles in half and finds the fold is always a line of symmetry, then concludes this holds for every isosceles triangle. This is an example of:

**a.** Deductive reasoning

**b.** Inductive reasoning

**c.** Circular reasoning

**d.** A logical fallacy

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Circles

What is the contrapositive of the statement: "If Two triangles are congruent, then their corresponding sides are equal."?

**a.** If Their corresponding sides are equal, then two triangles are congruent.

**b.** If Two triangles are not congruent, then their corresponding sides are equal.

**c.** If Their corresponding sides are not equal, then two triangles are not congruent.

**d.** If Two triangles are not congruent, then their corresponding sides are not equal.

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**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Area, Surface Area & Volume

What is the contrapositive of the statement: "If Two lines are perpendicular, then they form a right angle."?

**a.** If Two lines are not perpendicular, then they do not form a right angle.

**b.** If They do not form a right angle, then two lines are not perpendicular.

**c.** If Two lines are not perpendicular, then they form a right angle.

**d.** If They form a right angle, then two lines are perpendicular.

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Transformations & Symmetry

Angles $40^\circ$ and $60^\circ$: third angle:

**a.** 100

**b.** 90

**c.** 70

**d.** 80

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Foundations & Reasoning

Isosceles triangle with two sides $7$ and base $10$: perimeter:

**a.** 21

**b.** 17

**c.** 24

**d.** 26

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